

IMPACT OF DYNAMIC LINE RATING ON THE GERMAN ELECTRICAL ENERGY SYSTEM

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ABSTRACT

The deployment of new renewable energy sources remote from load centers leads to an increasing need for grid expansion. A notable example can be found in Germany where most wind energy is generated in the north and has to be transmitted to the industrial south [1]. In contrast to a strong increase in wind power capacity, planned by the current government, only some grid expansion projects are in the planning or implementation. Moreover most of this projects have been delayed due to administrative problems and protest activities [2]. This trend will likely narrow the bottleneck of the transmission system which can lead to more grid instabilities and curtailment [3].

The transmission capacity of an Overhead Transmission Line (OTL) directly depends on its ampacity. Traditionally, OTL ampacity is calculated at unfavorable, static weather conditions [3] referred to as Static Line Rating (SLR). By design, it underestimates the ampacity of an OTL and, when implementing in the real world, leads to an overestimated transmission capacity. In contrast, Dynamic Line Rating (DLR) considers time varying weather conditions to calculate OTL ampacity. Besides an increased cost-efficiency its implementation particularly enables a better wind power integration [4]. In literature, case studies consider DLR in simulations [3] or in small scale Energy System Optimization Models (ESOM) [5] [6] [7]. However, no case study evaluates the impact of DLR in a higher scale ESOM e.g. national ESOM.

In this paper, we analyze the impact of DLR on the German Electrical Energy System (EES). We build an ESOM based on the PyPSA-Eur framework [8]. The model is hourly resolved and considers a time horizon of one year. To calculate DLR in our model, we use the IEEE standard [9] with highly resolved historical weather data using atlite [10]. With our model, we compare the German EES using SLR and DLR. For the comparison, we analyze wind curtailment, the grid congestion as well as the generator deployment of both energy systems. This allows us to illustrate the benefits of DLR in the German EES.

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